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Machine Learning and Inductive Inference

Hendrik Blockeel
Sieben Bocklandt

Master of Artificial Intelligence
and multiple other Master programs

This academic year…

- 10 lectures; recordings made available.
  - Not always same location: check schedule
- Exercise sessions:
  - Exercises made available on Toledo, with solutions (separately)
  - Try to make them yourself, then check your solution
  - If still questions: come to one of the exercise sessions
- There are multiple parallel series of sessions (A-F). Please register for one series: see Toledo, "**course information**" / "**registration for exercise sessions**"

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Lecture 1 : Introduction

Questions:

What, according to you, is “machine learning”?

What can you use it for?

What does “inductive inference” mean?

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The RoboSail project

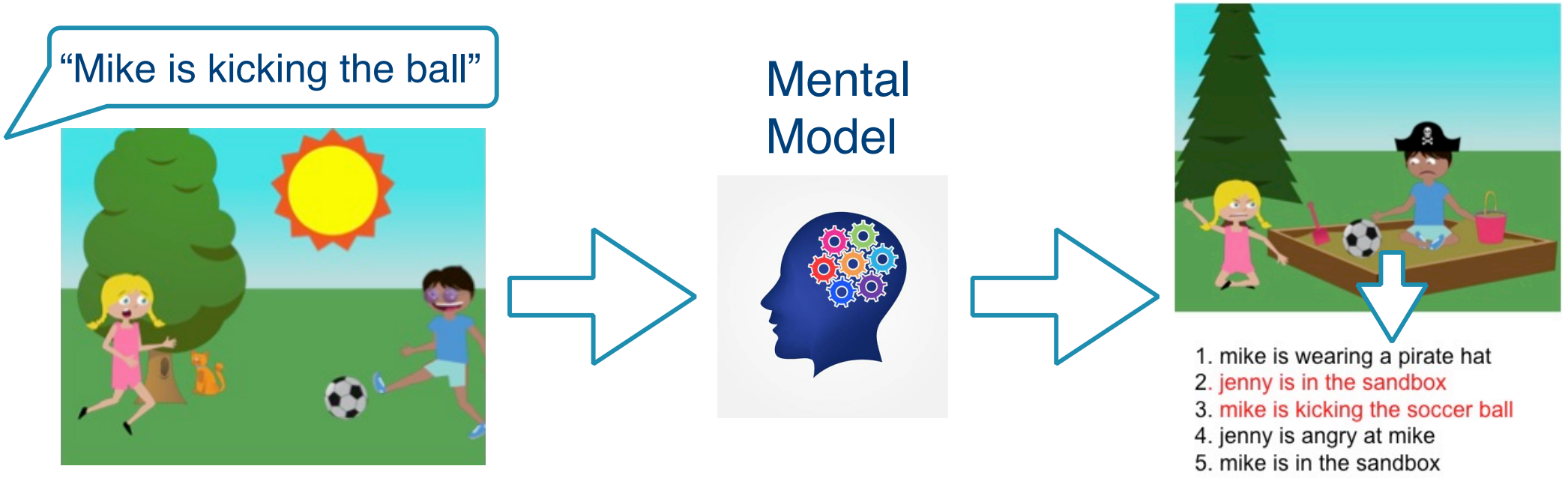
- Pieter Adriaans (Univ. of Amsterdam), around 2003: *first auto-pilot for sailing boats* (www.robosail.com)
  - No suitable mathematical theory => had to *learn* how to sail
  - Boat full of AI technology (agents, sensors, ...), including machine learning components



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Language learning

- Children learn language by simply hearing sentences being used in a certain context
- Can a computer do the same: given examples (sentence + description of context), learn the meaning of words/sentences?



Dataset: Zitnick et al., 2013

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Autonomous cars

- Very first race with fully autonomous cars was in… Censored
- DARPA grand challenge : have autonomous cars race each other on desert roads
- In 2004, no winner - “best” car got about 12 km far
- In 2005, five cars made it to the finish (212 km)





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The Robot Scientist

- King et al., Nature, 2004
- Scientific research, e.g., in drug discovery, is iterative:
  - Determine what experiment to perform
  - Perform the experiment
  - Interpret the results

Feedback

- Robot Scientist removes the human from the loop, by reasoning about its own learning process: which new experiments will be most informative?





2nd version, “Eve” (2015) discovered lead against malaria on first run

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Automating manual tasks

- E.g.: nurse rostering in a hospital: need to accommodate non-obvious constraints (e.g., leave enough time between shifts)
- Hard to automate, unless constraints can be *learned* from earlier examples

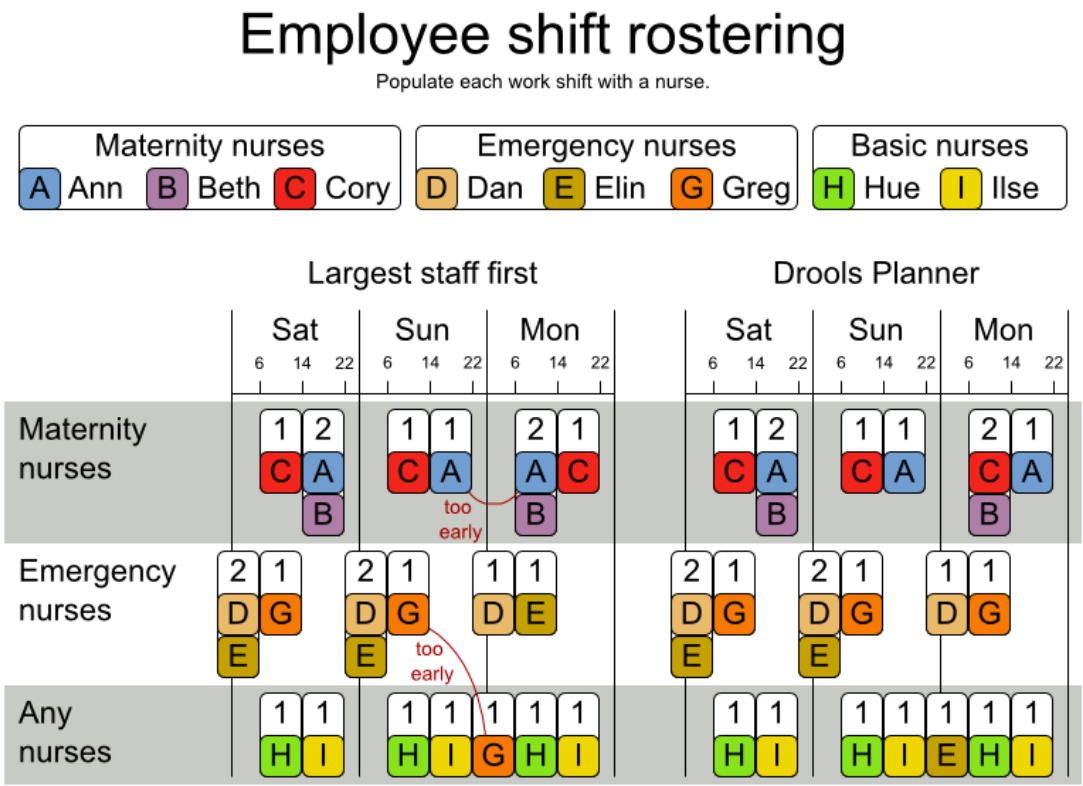


Illustration from L. De Raedt's SYNTH project,
picture by G. De Smet

Other applications…

- Recommender systems
  - Google, Facebook, Amazon, … try to show you ads you might like
- Email spam filters
  - By observing which mails you flag as “spam”, try to learn your preferences
- Natural language processing
  - Sentiment analysis: is this written review mostly positive or negative?
- … and many, many more



The Master Algorithm (P. Domingos) provides an excellent account of how machine learning affects our daily life

Definitions of machine learning?

- Tom Mitchell, 1996: Machine learning is the *study of how to make programs improve their performance on certain tasks from (own) experience*
  - "performance" = speed, accuracy, …
  - "experience" = earlier observations
- "Improve performance" in the most general meaning: this includes learning *from scratch.*
- Useful (in principle) for anything that we don't know how to program - computer "programs itself"
  - Vision: recognizing faces, traffic signs, …
  - Game playing, e.g., AlphaGo
  - Using natural language: conversations, translating, …
- Link to artificial intelligence : computer solves hard problems autonomously

Machine learning vs. other AI

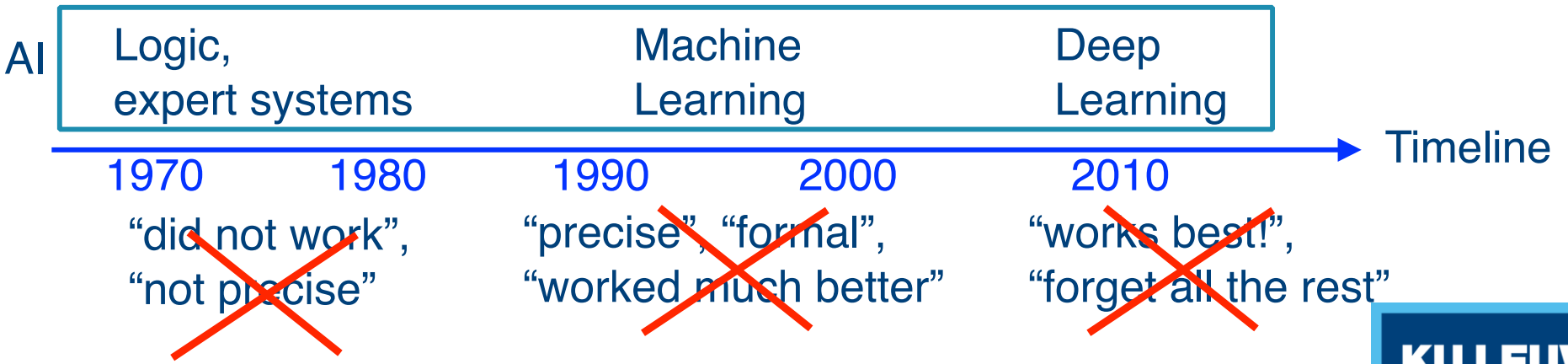
- In machine learning, the key is **data**
  - Examples of questions & their answer
  - Observations of earlier attempts to solve some problem

- "Inductive inference" = reasoning from specific to general
  - Statistics: sample → population
  - Philosophy of science: concrete observations → general theory

- This aspect of machine learning links it to data mining, data analysis, statistics, …

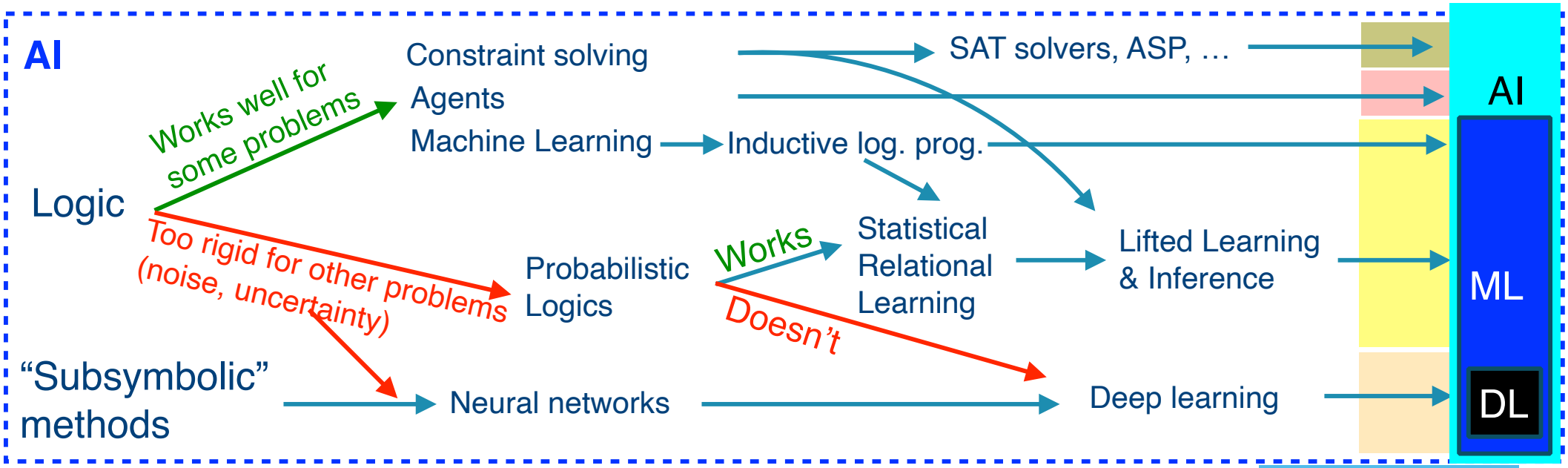
Machine learning and AI

- Many misconceptions about machine learning these days
- In the recent 5 years or so, "*Deep Learning*" has received a lot of attention: it revolutionized computer vision, speech recognition, natural language processing
- Avalanche of new researchers drawn to the field, without knowledge of the broader field of AI, or history of ML ("AI = ML = deep learning")
- See, e.g., A. Darwiche, https://www.youtube.com/watch?v=UTzCwCic-Do (also published in Communications of the ACM, October 2018)



Machine learning and AI

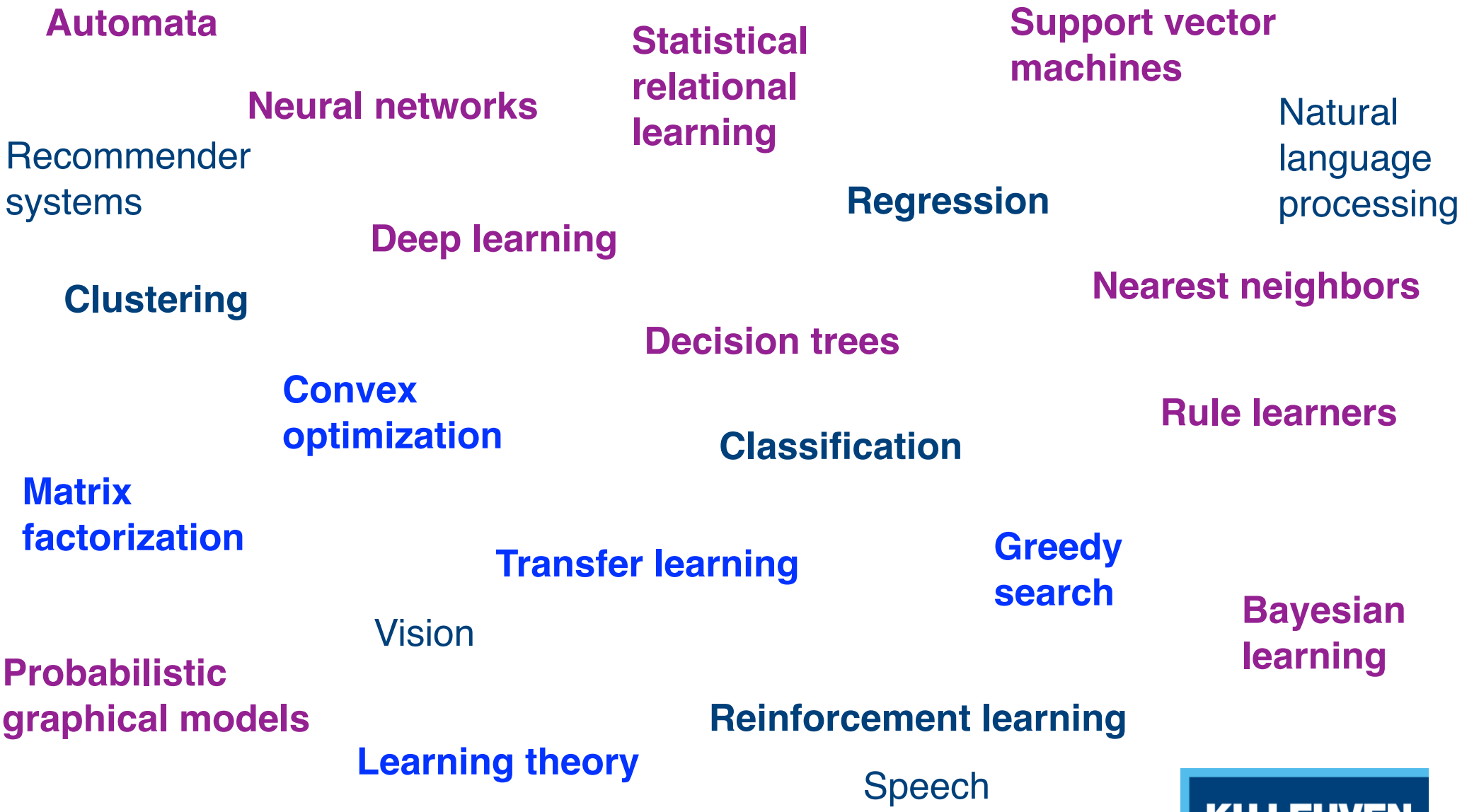
- My personal view on this : there is *still* progress on *all* fronts, deep learning is just one of them
- This course reflects that viewpoint
- (schema below is incomplete, just to illustrate complexity of scientific impact)



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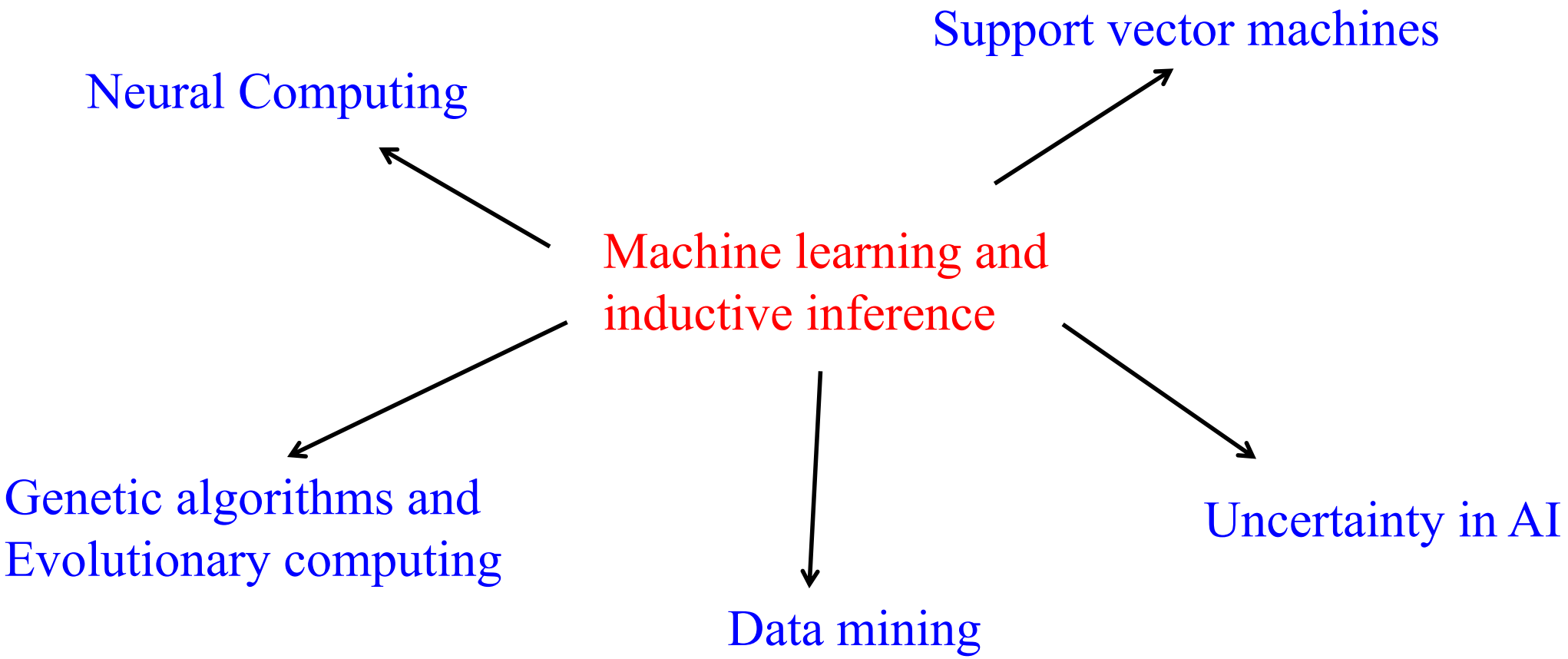
Tasks Techniques Models Applications

The machine learning landscape



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Related courses



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About this course…

- Primary goal of the course: provide an **overview** of machine learning, and **insight** in how the different methods work
- Secondary goal: enable you to **apply** machine learning

- 10 plenary lectures + 5 exercise sessions in smaller groups
  - Same session held several times per week
  - Only meant for answering questions, no full explanation of solutions

- ***Optional* c**ourse text (*Machine learning and inductive inference, H. Blockeel*) available at Acco bookshop
- Lectures slides and exercises available on Toledo

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About the exam…

- Written exam, closed book (formula sheet provided)
- Focus on insight, understanding, and the ability to *reason* about methods and *simulate* how they work
- Both theory and exercises

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Some basic concepts and terminology



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Machine learning

- ML in its typical form:
  - Input = dataset
  - Output = some kind of model
- ML in its most general form:
  - input = knowledge
  - output = a more general form of knowledge
- Learning = inferring general knowledge from more specific knowledge (observations ➔ model) = *inductive* inference

- Learning methods are often categorized according to the format of the input and output, and according to the goal of the learning process (but there are many more dimensions along which they can be categorized)

Example: Iris dataset

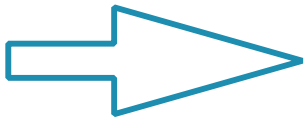
Sepal Length (cm)	Sepal Width (cm)	Petal Length (cm)	Petal Width (cm)	Species
5.1	3.5	1.4	0.2	setosa
4.9	3.0	1.4	0.2	setosa
4.7	3.2	1.3	0.2	setosa
7.0	3.2	4.7	1.4	versicolor
6.4	3.2	4.5	1.5	versicolor
6.9	3.1	4.9	1.5	versicolor
6.3	3.3	6.0	2.5	virginica
5.8	2.7	5.1	1.9	virginica
7.1	3.0	5.9	2.1	virginica

Learning task: learn a model that, given a description of a flower (length & width of sepals and petals), tells us what species of iris this is. I.e., learn a mathematical function f such that $f(SL, SW, PL, PW) = S$.

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Example: classifying pictures





"Iris Versicolor"

Learning task: learn a model that, given a *picture* of a flower, tells us what species of iris this is.

- Method 1: estimate SL, SW, PL, PW from picture, then use $f(SL, SW, PL, PW) = S$.
- Method 2: "end-to-end": $f(P) = S$, where P is a data structure that represents the entire picture (e.g., an array of pixels)

"Classic" methods

"Deep learning" methods

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Example: generating pictures

"Iris Versicolor"



Learning task: learn a model that, given a textual description, outputs a picture that fits that description.

$f(S) = P$, where P is a data structure that represents a picture and S is a textual description.

"Generative AI"

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Predictive versus descriptive

- **Predictive learning** : learn a model that can predict a particular property / attribute / variable from inputs
  - Many tasks are special cases of predictive learning
  - E.g., face recognition: given a picture of a face, say who it is
  - E.g., spam filtering: given an email, say whether it's spam or not

Name of task	*Learns a model that can …*
Concept learning / Binary classification	***Distinguish** instances of class C from other instances*
Classification	*Assign a **class** C (from a given set of classes) to an instance*
Regression	*Assign a **numerical** value to an instance*
Multi-label classification	*Assign a **set** of labels (from a given set) to an instance*
Multivariate regression	*Assign a **vector** of numbers to an instance*
Multi-target prediction	*Assign a **vector** of values (numerical, categorical) to an instance*
Ranking	*Assign **≤ or >** to a **pair** of instances*

Predictive versus descriptive

- **Descriptive learning** : given a data set, describe certain patterns in the dataset (or in the population it is drawn from)
- E.g., analyzing large databases:
  - “Bank X always refuses loans to people who earn less than 1200 euros per month”
  - “99.7% of all pregnant patients in this hospital are female”
  - “At supermarket X, people who buy cheese are twice as likely to also buy wine”

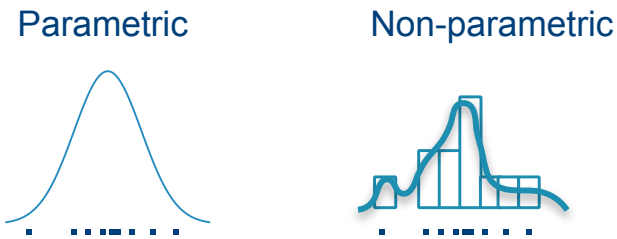
Function learning

- **Task** : learn a function X →Y that fits the given data (with X and Y sets of variables that occur in the data)
- Such a function will obviously be useful for predicting Y from X
- May also be descriptive, if we can understand the function
- Often, some **family of functions F** is given, and we need to estimate the parameters of the function f in F that best fits the data
  - e.g., linear regression : determine a and b such that $y = ax + b$ fits the data as well as possible
- What does "fit the data" mean? Measured by a so-called **loss function**
  - e.g., quadratic loss: $\sum_{(x,y)\in D} (f(x) - y)^2$ with f the learned function and D the dataset

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Distribution learning

- **Task:** given a data set drawn from a distribution, estimate this distribution
- Often made distinction: *parametric* vs. *non-parametric*
  - Parametric: a family of distributions is given (e.g., “Gaussian”), we only need to estimate the parameters of the target distribution
  - Non-parametric: no specific family is assumed
- Often made distinction: *generative* vs. *discriminative*
  - Generative: learn the joint probability distribution (JPD) over all variables (once you have that, you can *generate* new instances by random sampling from it)
  - Discriminative: learn a conditional probability distribution of Y given X, for some given set of variables X (called input variables) and Y (called target variables)



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These categorizations are somewhat fuzzy…

- A descriptive pattern may be useful for prediction
  - “Bank X always refuses loans to people who earn less than 1200 euros per month” (description)
  - Bob earns 1100 euros per month => Bank X will not give him a loan
- While functions are directly useful for prediction, a probability distribution can be used just as well
  - Given known information X, predict as value for Y, the value with the highest conditional probability given X

Parametric vs. non-parametric

- Parametric: a family of functions (or distributions, or …) is given, and each function is uniquely defined by the values of a fixed set of parameters
  - e.g. (function learning): linear regression
  - e.g. (distribution learning): fitting a gaussian
- Non-parametric: no specific family of functions is assumed
  - Typically, we are searching a space that contains models with varying structure, rather than just different parameter values
  - This often requires searching a discrete space
  - E.g.: decision trees, rules, …. (see later)

Link with “explainable AI”

- **Explainable AI** (**XAI**) refers to the study of AI systems that can explain their decisions / whose decisions we can understand
- Two different levels here:
  - We understand the (learned) model used for decision making
  - We understand the individual decision
- E.g. “I could not get a loan because I earn too little”: we can understand this decision even if we don’t know the whole decision process the bank uses
- A learned model that is not straightforward to interpret, is called a **black-box** model
- Machine learning poses additional challenges for XAI, as it often learns black-box models

Responsible AI : challenges

- **Privacy-preserving** data analysis
  - We need lots of data to learn from; this may include personal data
  - How can we guarantee that the analysis of these data will not violate the privacy of the people described by the data?
  - When data is collected, consent is needed for a specific purpose, and data can only be used for that purpose. How can we guarantee it won't be abused?

- Learning "**safe**" models : models that will not violate certain constraints that are imposed (including constraints on bias, discrimination, privacy, …)

Predictive learning

- A very large part of machine learning focuses on predictive learning
- In the following, we introduce some more concepts and terminology in that setting

Prediction: task definition

The prediction task, in general:

- Given: a description of some instance
- Predict: some property of interest (the “target”)

Examples:

- classify emails as spam / non-spam
- classify fish as salmon / bass
- forecast tomorrow’s weather based on today’s measurements

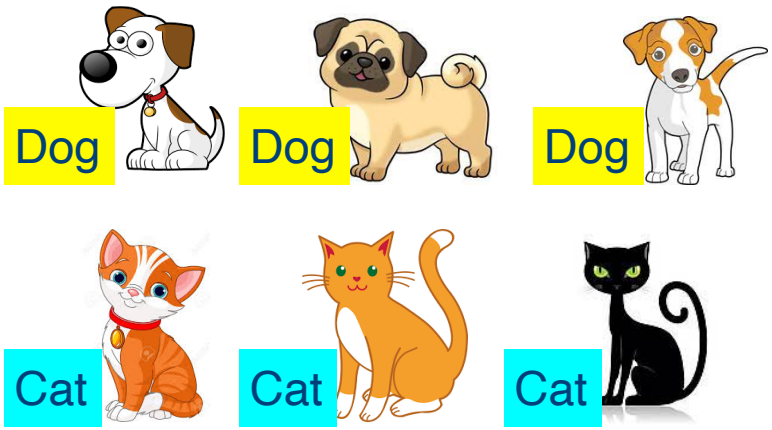
How? By analogy to cases seen before

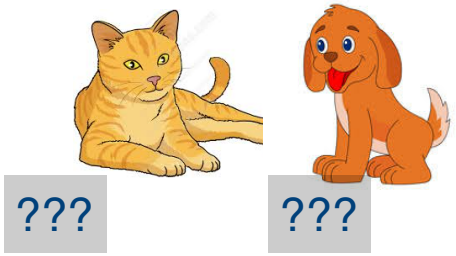
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Terminology

- **Training set**: a set of **examples**, instance descriptions that include the target property (a.k.a. **labeled** instances)
- **Prediction set**: a set of instance descriptions that do *not* include the target property (“unlabeled” instances)

- Prediction task : predict the labels of the unlabeled instances





Inductive vs. transductive learning

We can consider as outcome of the learning process, either

- the predictions themselves: **transductive learning**
- or: a function that can predict the label of any unlabeled instance: **inductive learning**



Transduction: outcome=*predictions*

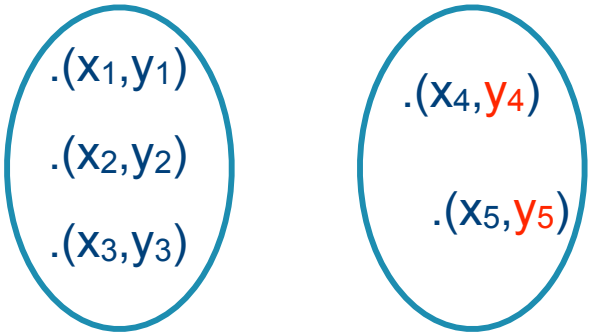
Induction: outcome = *function for making predictions*

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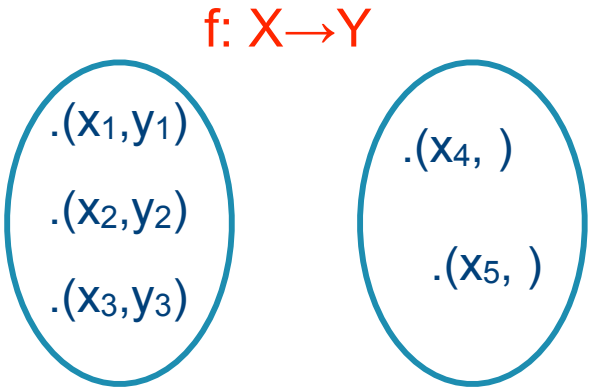
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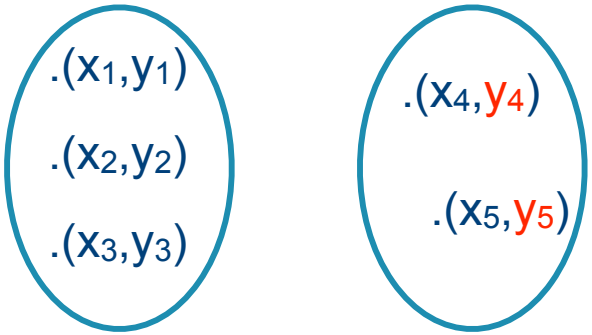
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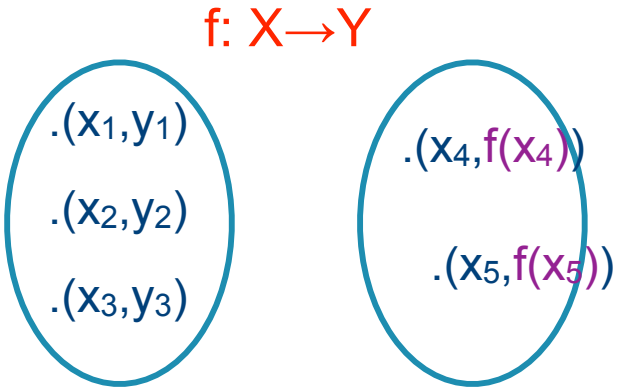
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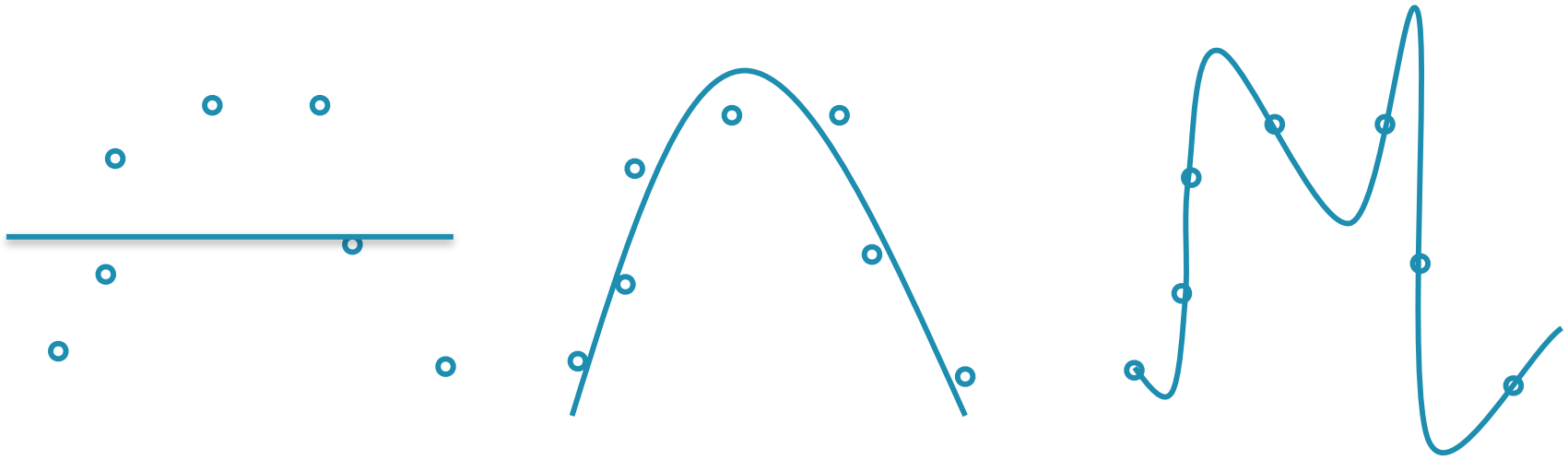


Induction: outcome = *function for making predictions*

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Overfitting and underfitting

- “Occam’s razor”: among equally accurate models, choose the simpler one
- Trade-off: explain data vs. simplicity
- Both overfitting and underfitting are harmful



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Levels of supervision

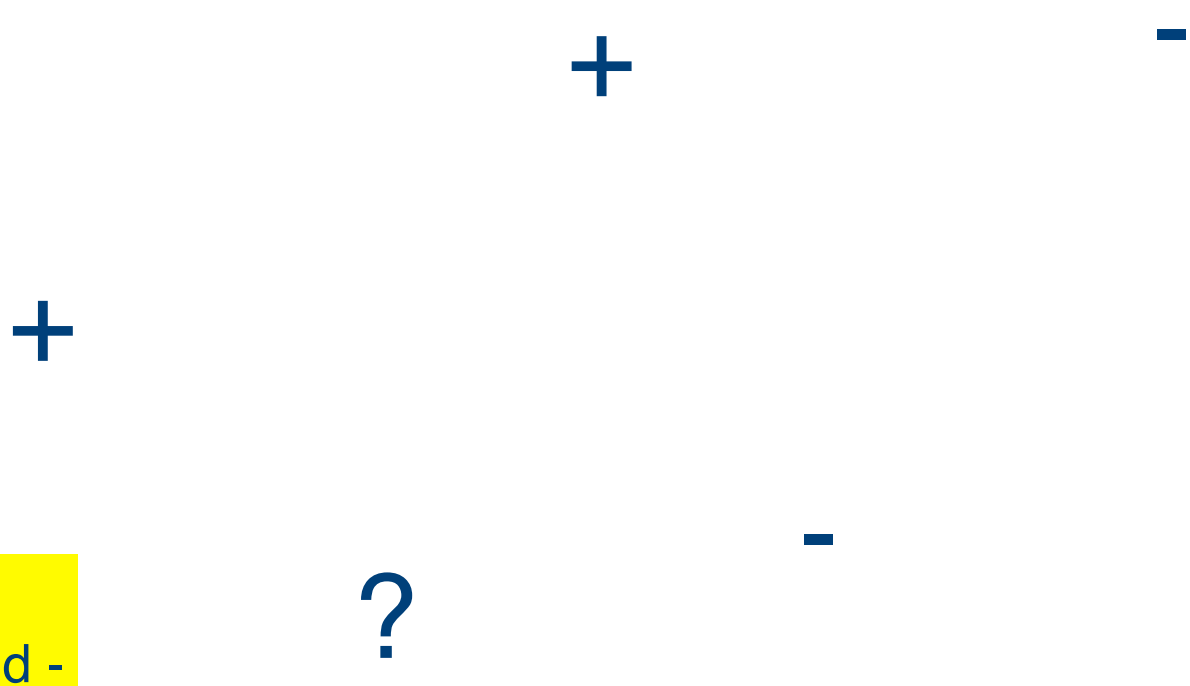
- **Supervised learning**: learning a (predictive) model from *labeled* instances (as in cats & dogs example)

- **Unsupervised learning**: learning a model from *unlabeled* instances
  - such models are usually not directly predictive (without *any* information on what to predict, how could you learn from that?)
  - still useful indirectly, or for non-predictive tasks: see later

- **Semi-supervised learning**: learn a predictive model from *a few labeled* and *many unlabeled* examples

Semi-supervised learning

- How can unlabeled examples help learn a better model?

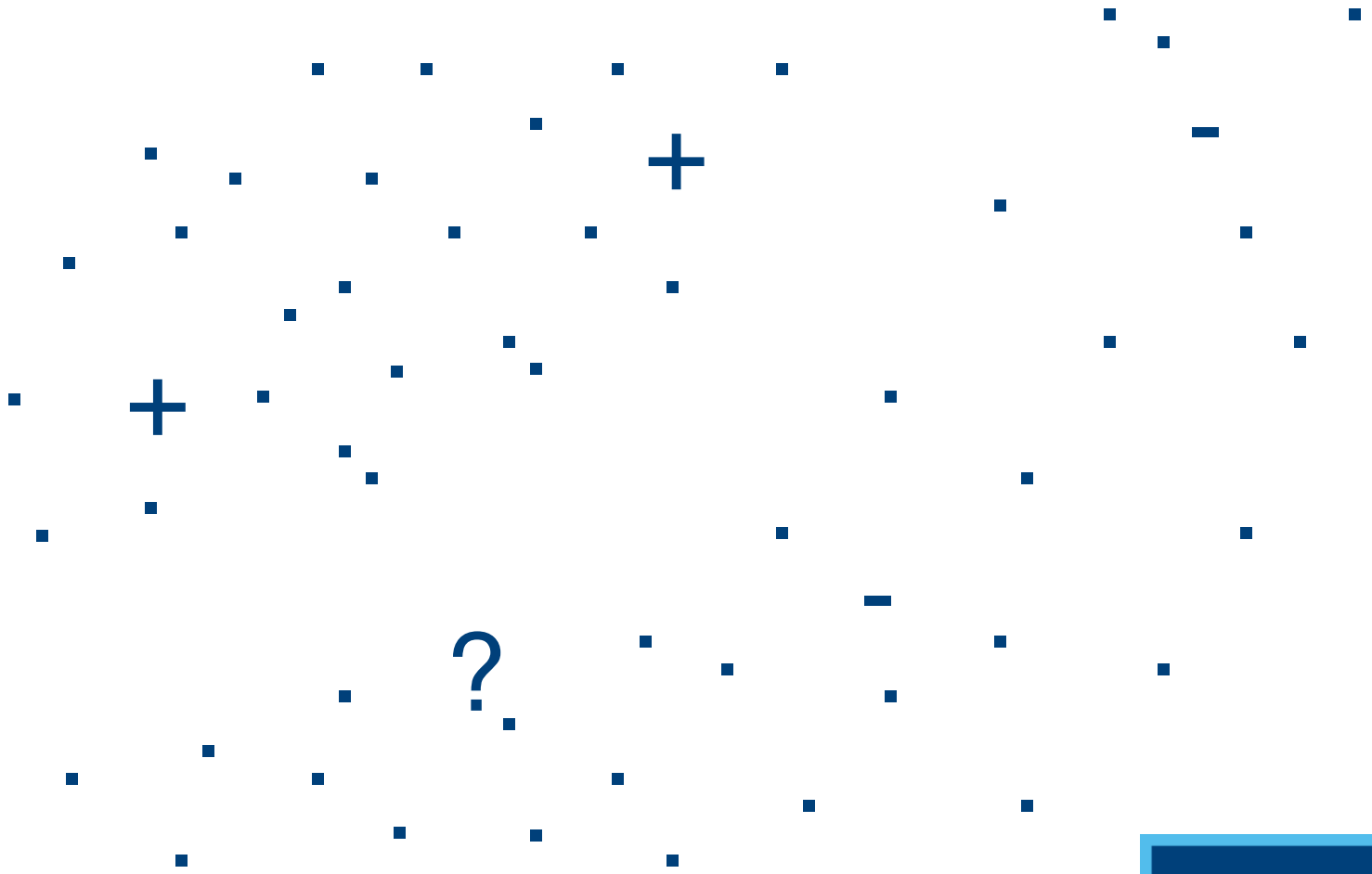


This illustration:
- 2 classes, called + and -
- Representing instances in a 2-dimensional space

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Semi-supervised learning

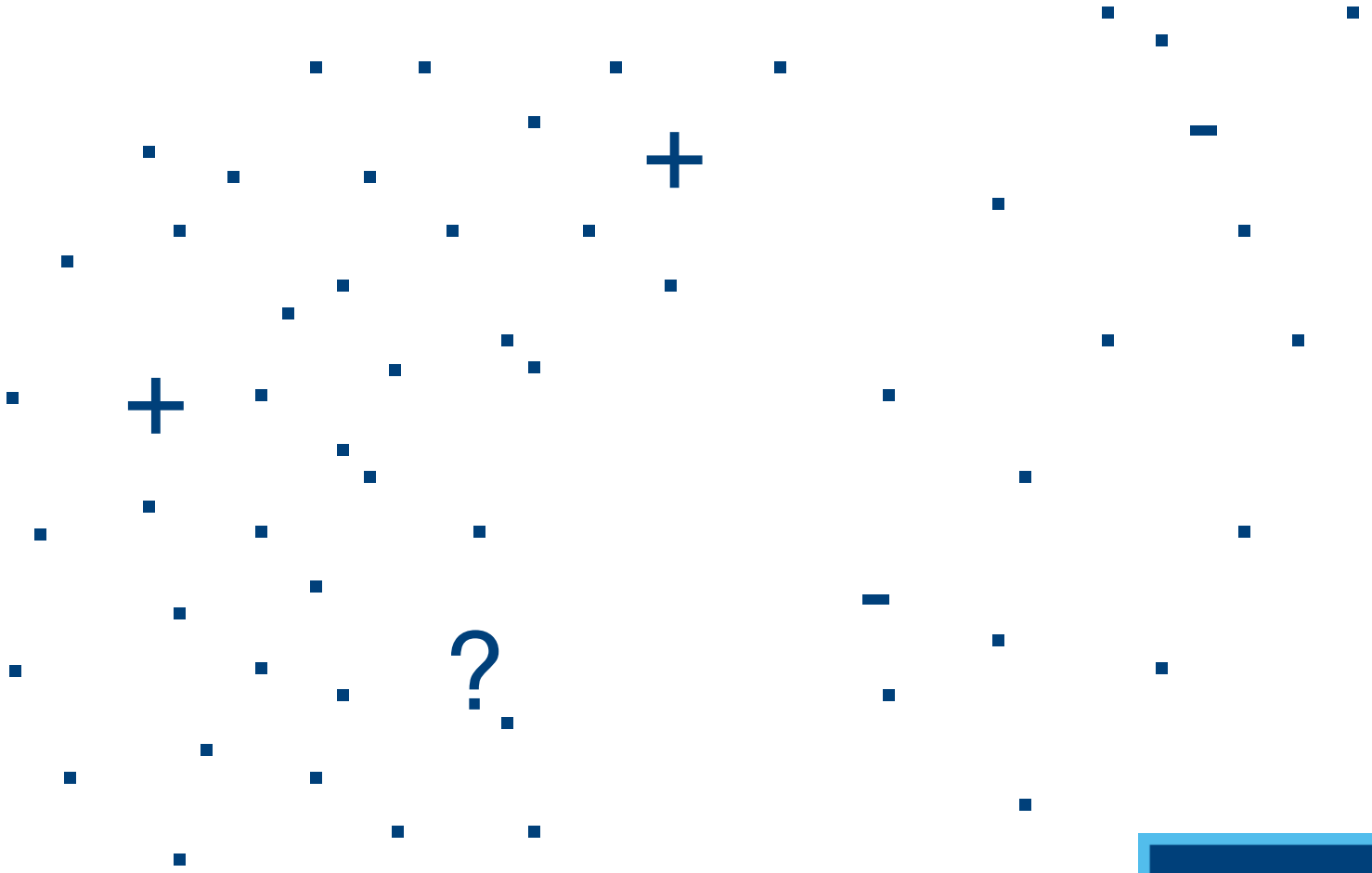
- How can unlabeled examples help learn a better model?



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Semi-supervised learning

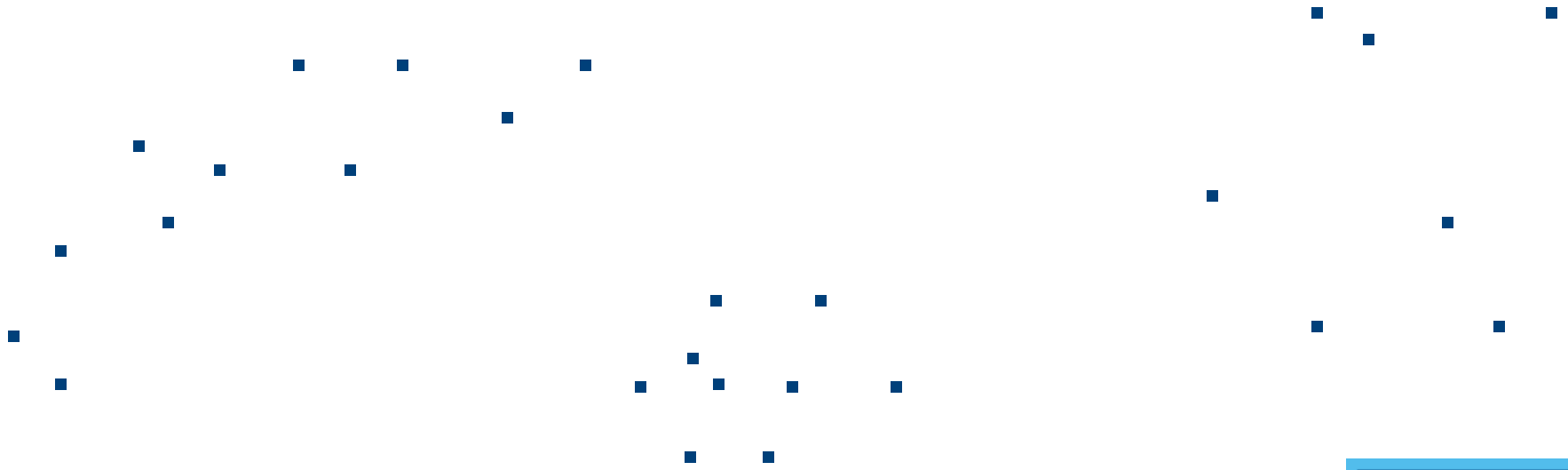
- How can unlabeled examples help learn a better model?



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Unsupervised learning

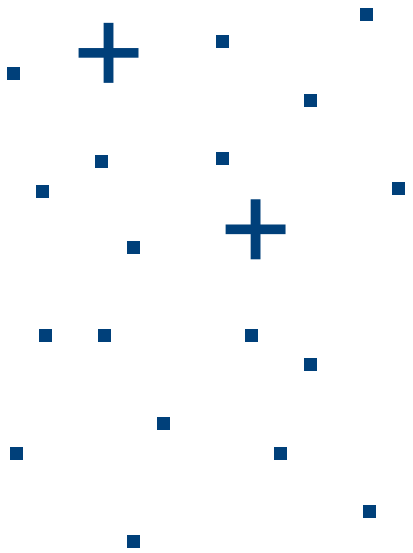
- Can you see three classes here?
- Even though we don't know the *names* of the classes, we still see some structure (*clusters*) that we could use to predict which class a new instance belongs to
- Identifying this structure is called *clustering*
- From a predictive point of view, this is unsupervised learning



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PU-learning

- PU-learning is a special case of semi-supervised learning
- PU stands for “positive / unlabeled”
- All the labeled examples belong to one class (called the “positive” class)



Learning the meaning of “kicking the ball” requires PU-learning because: When Mike kicks the ball, the sentence *may* mention this, or not. When Mike does not kick the ball, it is never mentioned that he does not.

“Mike is kicking the ball”



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Weakly supervised learning

- Weakly supervised learning is a generalized form of semi-supervised learning
- Semi-supervised: for a single instance, we either know its label or we do not
- Weakly supervised: we may have partial information about a label
  - e.g., it is certainly a member of a given set (= **superset learning**)
  - e.g., at least one instance among a given set of instances has the label, but we do not know which one (= **multi-instance learning**)
  - e.g., we know two instances have the same label, but we don't know which one it is (= **constraint-based clustering**)
  - …

"This is either a Ferrari or a Lamborghini"

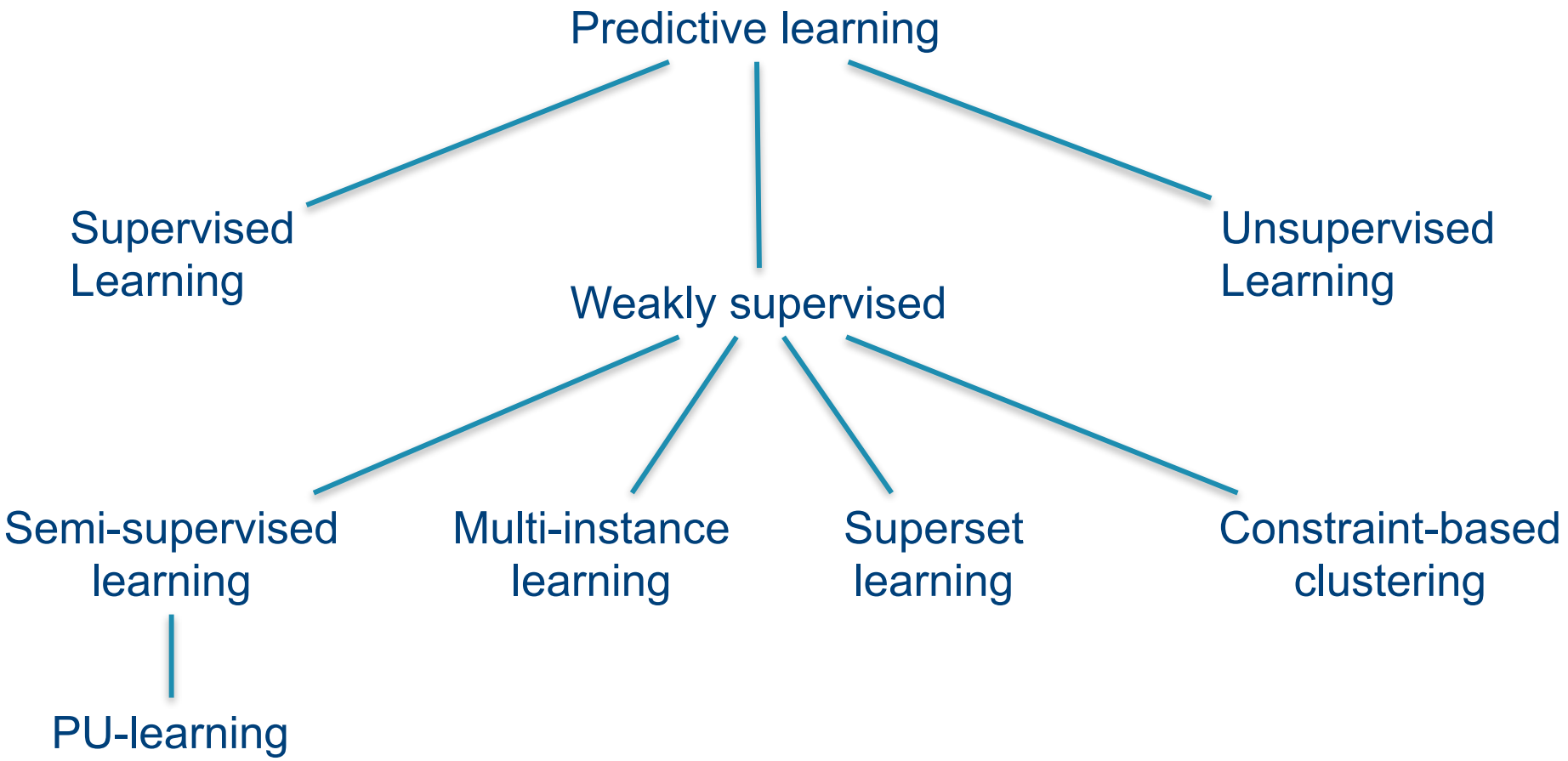


"There's a Lamborghini in this picture"



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Relationship between different supervision settings



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Reinforcement learning

- Learning setting: an agent can be in a number of different states. It can take actions, which change its state and may yield a reward. The agent's goal is to maximize its rewards in the long run.
  - Think of, e.g., chess: state = board state; action = make a move. You want to make moves that cause you to win in the long run
- The learning task is to learn a function f that, for any state s, predicts the best action to take. But *supervision is indirect*: the learner is not told what the best action is. Instead, it sees sequences (x_i, y_i, r_i) where
  - y_i is some action (not necessarily the best) that was taken in state x_i,
  - r_i is the corresponding reward, and
  - taking action y_i in state x_i led to state x_{i+1}
- The best action is not necessarily the one that leads to the highest immediate reward! An action can lead to states with higher reward opportunities later on
  - Cf. chess: you might want to sacrifice a piece to gain benefits in the long term
- This is a more complex learning setting that deserves its own study

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Format of input data

- Input is often assumed to be a *set* of instances that are all described using the same variables (features, attributes)
  - The data are “i.i.d.”: “independent and identically distributed”
  - The training set can be seen as a random sample from one distribution
  - The training set can be shown as a table (instances x variables) : *tabular data*
  - This is also called the **standard setting**

- There are other formats: instances can be
  - nodes in a graph
  - whole graphs
  - elements of a sequence
  - …

Format of input data: *tabular*

Training set

Sepal length	*Sepal width*	*Petal length*	*Petal width*	*Class*
5.1	*3.5*	*1.4*	*0.2*	*Setosa*
4.9	*3.0*	*1.4*	*0.2*	*Setosa*
7.0	*3.2*	*4.7*	*1.4*	*Versicolor*
6.3	*3.3*	*6.0*	*2.5*	*Virginica*

Prediction set

Sepal length	*Sepal width*	*Petal length*	*Petal width*	*Class*
4.8	*3.2*	*1.3*	*0.3*	*?*
7.1	*3.3*	*5.2*	*1.7*	*?*

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Format of input data: *sequences*

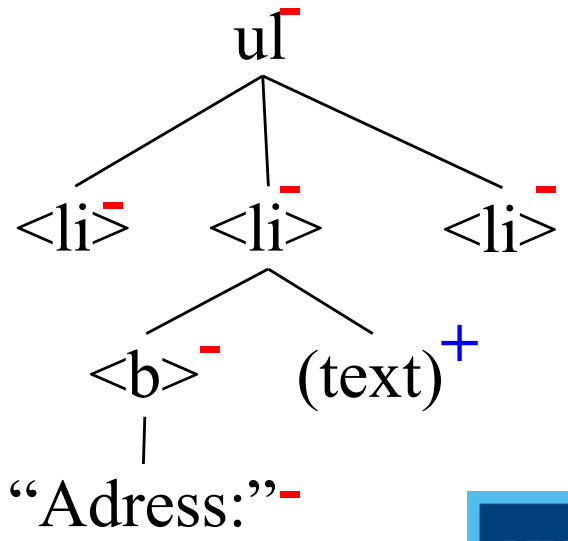
- Learning from sequences:
  - 1 prediction per sequence?
  - 1 prediction per element?
- 1 element in sequence can be …
  - A number (e.g., time series)
  - A symbol (e.g., strings)
  - A tuple
  - A more complex structure

```
abababab: +
aabbaabb: -
```

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Format of input data: *trees*

- 1 prediction per tree / per node in the tree
- Nodes can be …
  - Unlabeled
  - Labeled with symbols (e.g., HTML/XML structures)
  - …

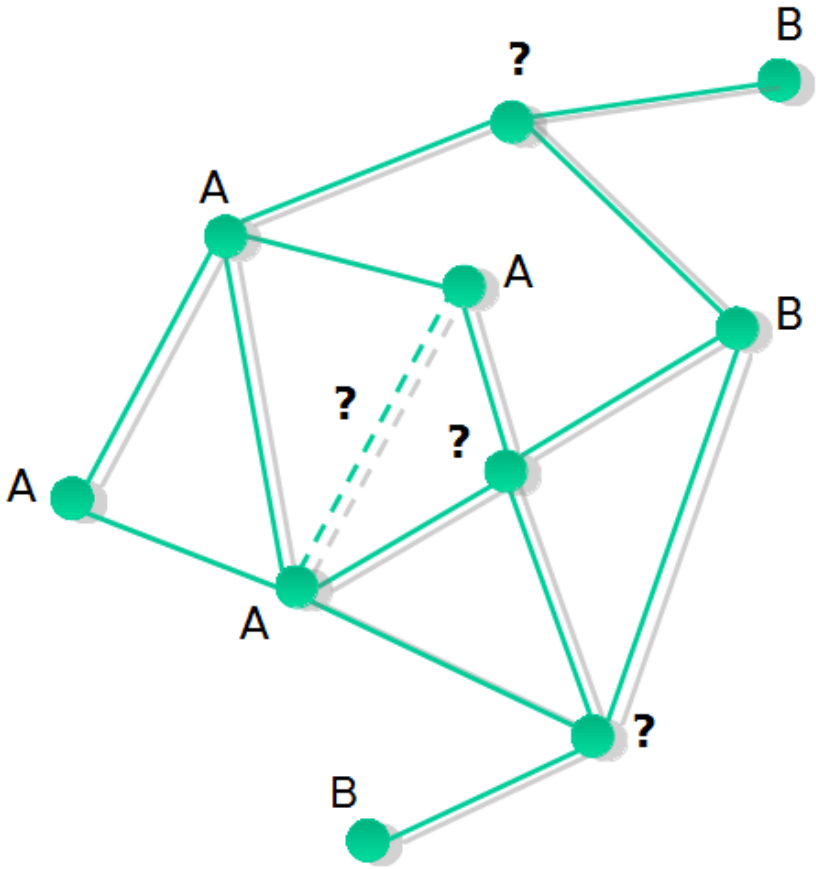


E.g.: this tree indicates a “positive” a text field preceded by **Address:** in a list context

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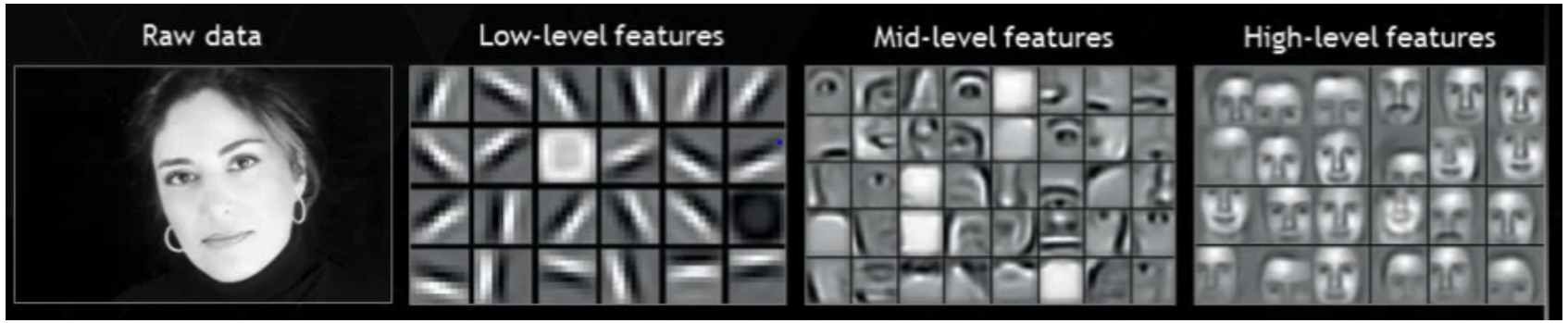
Format of input data: *graph*

- Example: Social network
- Target value known for some nodes, not for others
- Predict node label
- Predict edge
- Predict edge label
- …
- Use *network structure* for these predictions



Format of input data: *raw data*

- "Raw" data are in a format that seems simple (e.g., a vector of numbers), but components ≠ meaningful features
- Example: photo (vector of pixels)
- Raw data often need to be processed in a non-trivial way to obtain meaningful features; on the basis of these features, a function can be learned
- This is what deep learning excels at



(Image: Nielsen, 2017, *Neural networks and deep learning*)

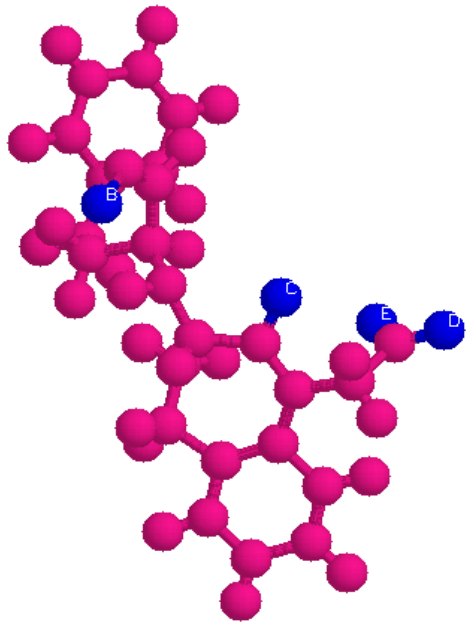
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Format of input data: *knowledge*

- "Knowledge" can consist of facts, rules, definitions, ….
- We can represent knowledge about some domain in a *knowledge representation* language (such languages are often based on logic)

```
atm(m1,a1,o,2,3.43,-3.11,0.04).
atm(m1,a2,c,2,6.03,-1.77,0.67).
...
bond(m1,a2,a3,2).
bond(m1,a5,a6,1).
bond(m1,a6,a7,du).
...
```

```
...
hacc(M,A):- atm(M,A,o,2,_,_,_).
hacc(M,A):- atm(M,A,o,3,_,_,_).
hacc(M,A):- atm(M,A,s,2,_,_,_).
hacc(M,A):- atm(M,A,n,ar,_,_,_).
zincsite(M,A):-
        atm(M,A,du,_,_,_,_).
hdonor(M,A) :-
                atm(M,A,h,_,_,_,_),
                not(carbon_bond(M,A)), !.
...
```



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Data preprocessing

- Data may not be in a format that your learner can handle
- **Data wrangling**: bring it into the right format
- Even if it's in a format you learner can handle (e.g., tabular), the features it contains may not be very informative, or there may be very few relevant features among many irrelevant ones.
  - E.g.: individual pixels in an image are usually not very informative
- **Feature selection**: select among many input features the most informative ones
- **Feature construction**: construct new features, derived from the given ones

What learning method to use?

- Which learners are suitable for your problem, depends strongly (but not solely!) on the structure of the input data
- Most learners use the *standard* format
  - A set of instances, where each instance is described by a fixed set of *attributes* (a.k.a. *features*, *variables*)
  - also called *attribute-value* format or *tabular* format
- At the other extreme, *inductive logic programming* handles any kind of knowledge that can be represented using clausal logic
  - This includes sequences, graphs, …

What learning method to use?

- The data format and the learning task impose strong constraints on which learning methods can be used
- Other aspects determine whether the method performs well:
  - Inductive bias (see later)
  - Ability to handle missing values
  - Ability to handle noise
  - Ability to handle high-dimensional data
  - Ability to handle large datasets
  - Ability to generalize from small datasets (avoid overfitting)
- We'll cover many of these aspects at different points in the course

Output formats, methods (overview)



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Output formats

- The output of a learning system is a *model*
- Many different types of model exist
- The learning algorithm or method is strongly linked to the type of model

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Different views of the landscape

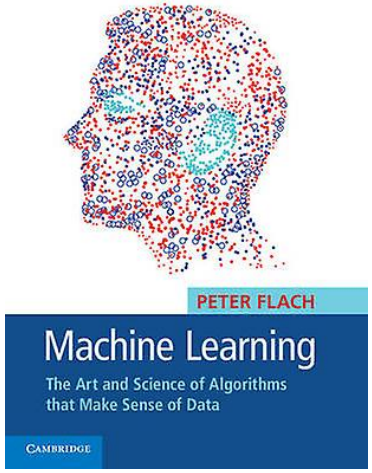
Domingos:
“five tribes”

- Symbolists
- Connectionists
- Evolutionaries
- Bayesians
- Analogizers



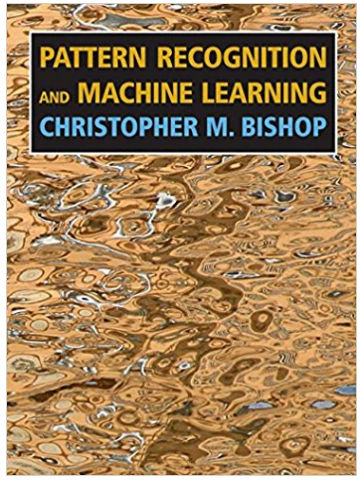
Flach:
“three types of models”

- Probabilistic
- Geometric
- Logical



Bishop:
“it’s all Bayesian”

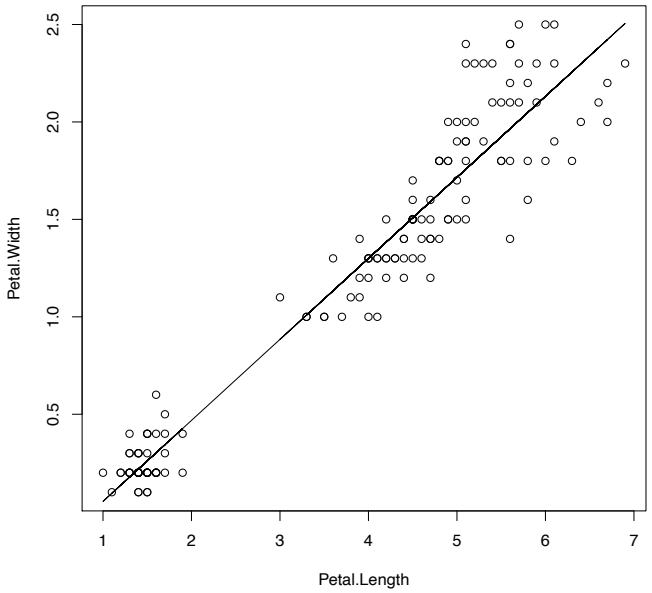
- Bayes
- Bayes
- Bayes



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Parametrized functions

- Typically, a certain format for the functions is provided; e.g.: linear functions of the inputs
- Within this set, we look for the parameter values that best fit the data
- Standard example: *linear regression*



$$y = ax + b$$

*width = 0.416*length - 0.363*

Conjunctive concepts

- A conjunctive concept is expressed as a set of conditions, all of which must be true
- “x has class C if and only if <condition1> *and* <condition2> *and* … *and* <condition k>”

- E.g.: accept application for mortgage if and only if :
  salary ≥ 3 * monthly payback ***and*** no other mortgage running

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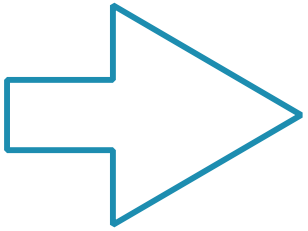
Rule sets

- A rule set is a set of rules of the form “if … then …” or “if … then … else …”
- Example: definition of leap years

Examples of leap years:
1900, 1992, 2004, …

Examples of non-leap years:
1993, 2000, 2011, 2018, …

Input



Output

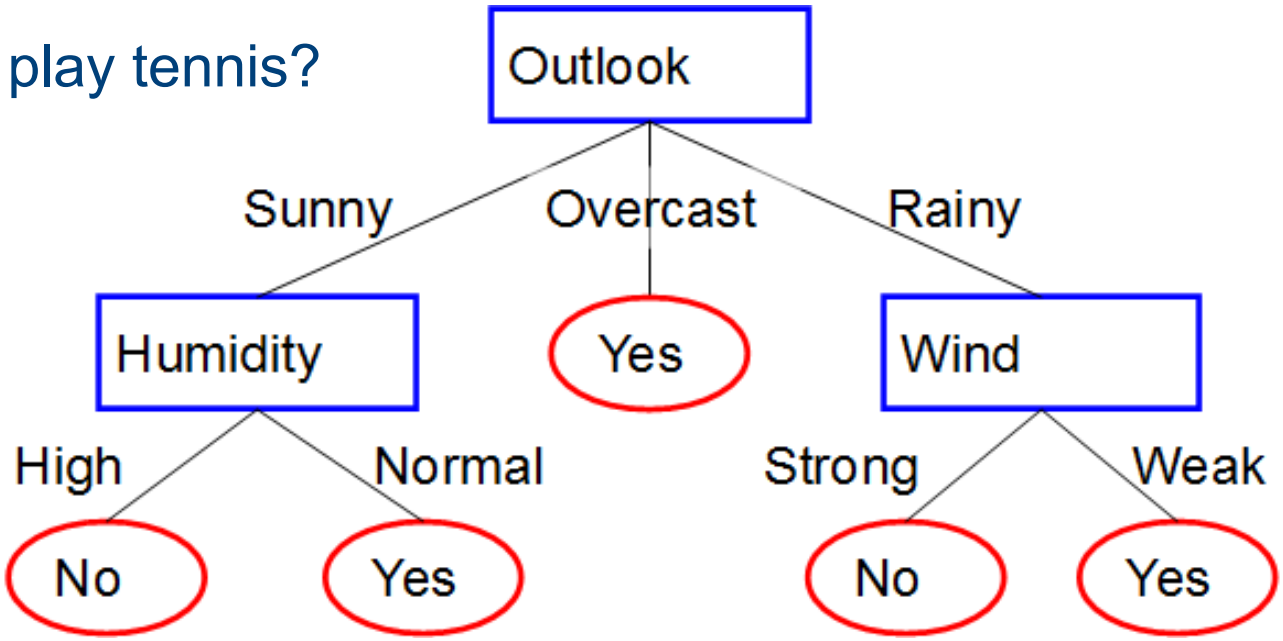
If year is a multiple of 400 **then** leap
else if year is a multiple of 100 **then** not leap
else if year is multiple of 4 **then** leap
else not leap

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Decision trees

- A decision tree represents a stepwise procedure to arrive at some decision

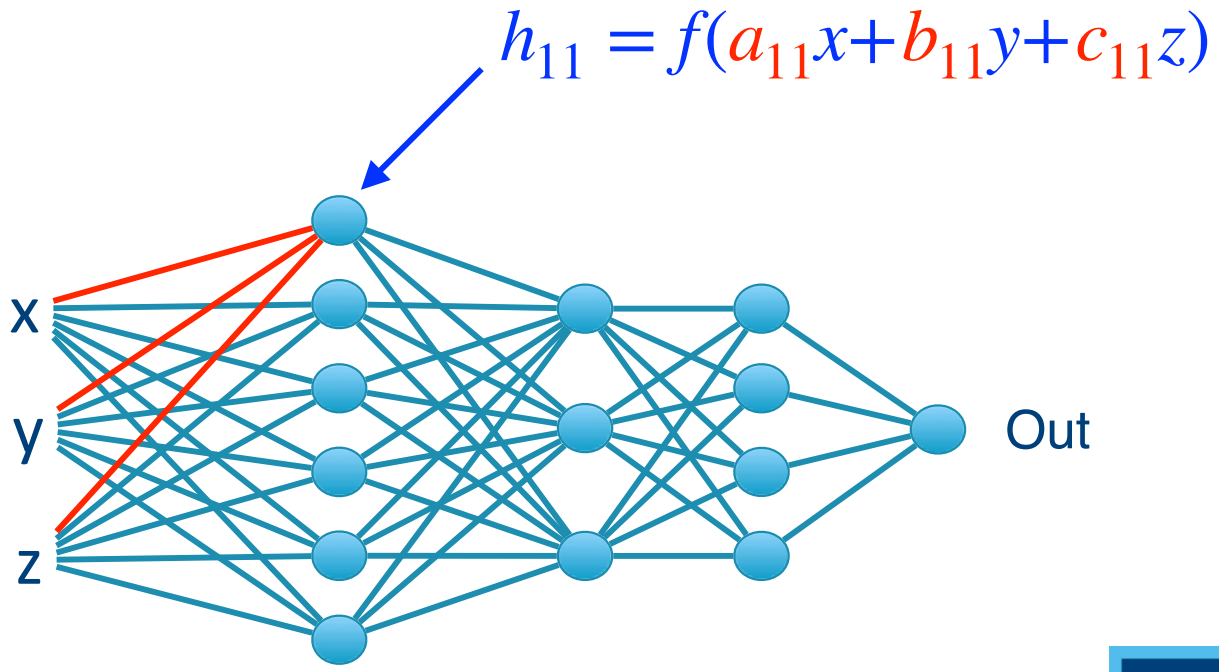
Is today a good day to play tennis?



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Neural networks

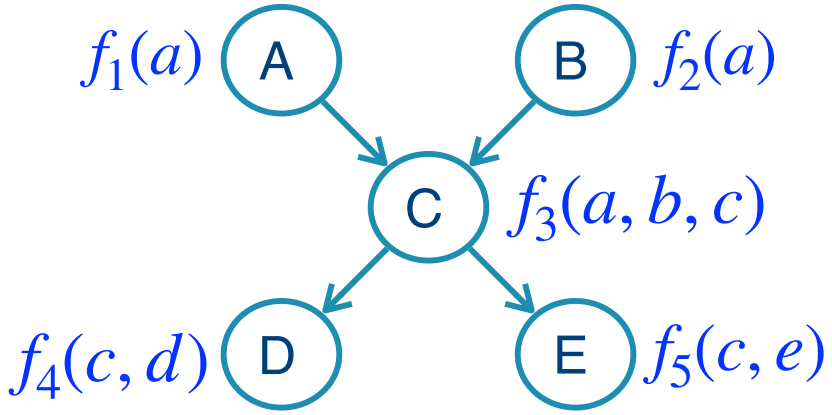
- A neural network is a complex structure of neurons, each of which aggregate multiple input signals into a single output signal



Probabilistic graphical models

- A PGM represents a (high-dimensional) joint distribution over multiple variables as a product of (low-dimensional) factors
- Different type of PGMs: Bayesian networks, Markov networks, factor graphs, …

Example: Bayesian network



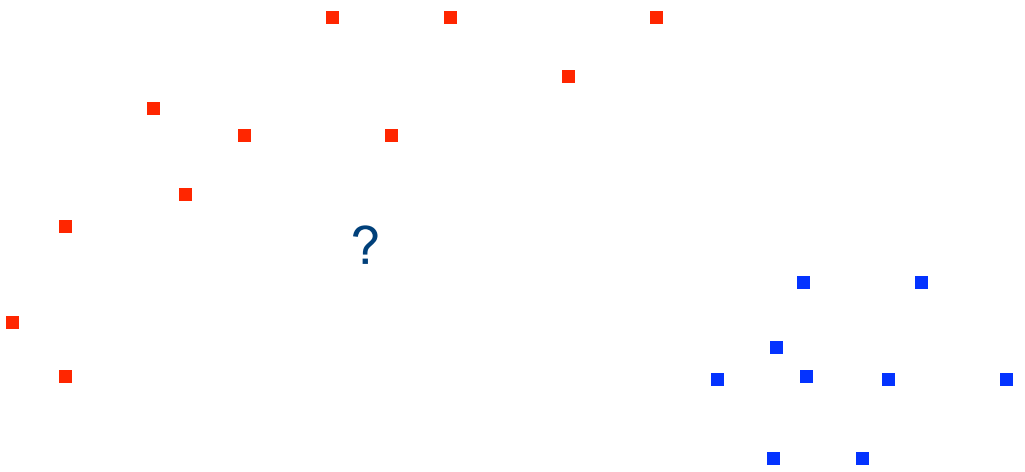
$$f(a,b,c,d,e) = f_1(a) \cdot f_2(b) \cdot f_3(a,b,c) \cdot f_4(c,d) \cdot f_5(c,e)$$

$$P(A,B,C,D,E) = P(A) \cdot P(B) \cdot P(C|A,B) \cdot P(D|C) \cdot P(E|C)$$

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Instance-based learning (a.k.a. “nearest neighbor methods”)

- The “model” is simply the data set itself
- Predictions for new cases are made by comparing it to earlier observed cases
- If it’s similar for observed features, it’s probably also similar for unobserved (to be predicted) features



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Coming lectures…

- In the rest of this course, we will have a closer look at many of these different methods
- Starting next week with a closer look at learning predictive functions and nearest neighbor methods

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